

Revisiting the Weerakoon–Fernando Variant of Newton's Method

Replication, Extensions, and a Critical Assessment of Convergence Order

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CSE 402 (Numerical Methods) — Project Presentation

1. The Problem and the Key Idea

Goal: solve $f(x) = 0$ (e.g. $\cos x - x = 0$).

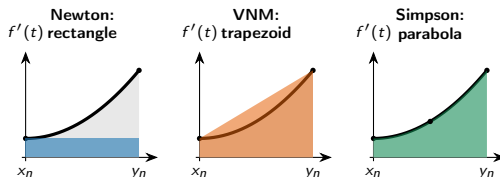
Newton's method (NM) — digits roughly *double* each step:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad e_{n+1} \approx C_2 e_n^2$$

Base paper (Weerakoon & Fernando, 2000):
replace the rectangle by a *trapezoid* $\Rightarrow$ **VNM**,
order **3**, no f'' needed:

$$y_n = x_n - \frac{f(x_n)}{f'(x_n)}, \quad x_{n+1} = x_n - \frac{2f(x_n)}{f'(x_n) + f'(y_n)}$$

$$e_{n+1} = (C_2^2 + \frac{1}{2}C_3)e_n^3$$



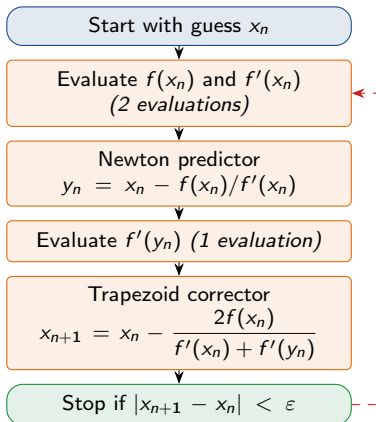
Each method estimates the **area under** $f'(t)$ between x_n and y_n .
The rectangle badly misses it; the trapezoid is much closer.

In plain words

Newton follows the tangent line down to the axis using *one* slope. VNM also measures the slope where that tangent lands (y_n), then **averages the two slopes** — a better travel direction, so it needs fewer steps.

2. The Methods Studied: Algorithm and Extensions

VNM algorithm (one iteration)



Three extensions we designed and tested

Method	Idea	Evals	Expected	Verified	$E = p^{1/m}$
NM	rectangle	2	2	2	1.414
VNM	trapezoid	3	3	3 ✓	1.442
S-VNM	Simpson parabola	4	4	3 ✗	1.316
St-VNM	no f' (Steffensen)	4	3	2 ✗	1.189
Multi-VNM	systems $F(X) = 0$	$1F+2J$	3	3 ✓	—

How each method was checked

- **18 test cases** from the base paper (replication)
- **6 functions + one 2×2 system** (double precision)
- **COC** = computational order of convergence, measured in **400-digit** arithmetic
- **Symbolic Taylor series** (SymPy) to derive the exact error equation

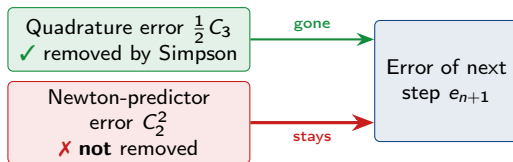
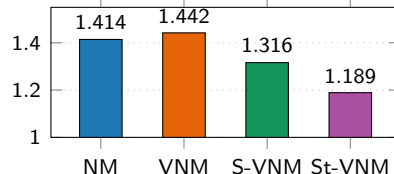
In plain words

E = speed per function evaluation (higher is better). **Red** = extension missed the order we hoped for (next slide).

3. Key Finding: Why the Extensions Missed Their Target Order

Method	Symbolic error equation (verified to 6 digits)
NM	$e_{n+1} = C_2 e_n^2$
VNM	$e_{n+1} = (C_2^2 + \frac{1}{2} C_3) e_n^3$
S-VNM	$e_{n+1} = C_2^2 e_n^3$ (still cubic!)
St-VNM	$e_{n+1} = \frac{1}{2} f'(\alpha) C_2 e_n^2$ (only quadratic)

Efficiency index E (higher = better)



In plain words

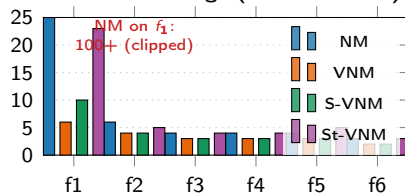
S-VNM: a better “area” rule (Simpson) fixes one error source, but the method still starts from Newton’s *predicted* point y_n , which is slightly wrong — that leftover error caps the order at 3. **St-VNM:** replacing f' by a finite-difference slope makes the slope itself inaccurate, dropping the order to 2. Result: **VNM** is still the best method of the family.

4. Experiment A: Benchmark on Six Functions (double precision)

Iterations / function evaluations (NFEV)

$f(x)$	x_0	NM	VNM	S-VNM	St-VNM
$x^3 + 4x^2 - 10$	-0.5	100+/200+	6/18	10/40	23/91
$\sin^2 x - x^2 + 1$	1.0	6/12	4/12	4/16	5/19
$e^x - 3x$	0.5	4/8	3/9	3/12	4/15
$\cos x - x$	0.5	4/8	3/9	3/12	4/15
$x^3 - 2x - 5$	2.0	4/8	3/9	3/12	5/19
$\ln x + x - 2$	1.5	3/6	2/6	2/8	3/11
Total (f_2 - f_6)		21/42	15/45	15/60	21/79
ACOC (typical)		2.00	3.0	3.0	2.00

Iterations to converge (fewer is better)

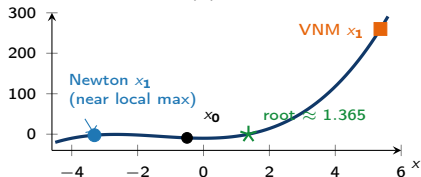


Take-aways

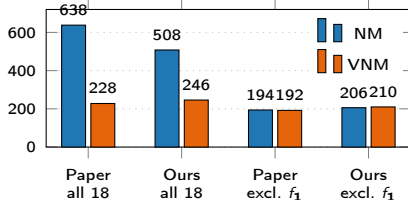
- Third-order methods (**VNM**, **S-VNM**) need **fewer iterations**; on easy functions VNM and NM use nearly the **same number of evaluations** (45 vs 42), as E predicts.
- VNM** shines on hard start f_1 , $x_0 = -0.5$: **6 iterations vs. 100+** for Newton.

5. Replication: Where VNM's Advantage Really Comes From

Why Newton wanders: $f_1(x) = x^3 + 4x^2 - 10$, $x_0 = -0.5$



Total NFEV over the 18 test cases

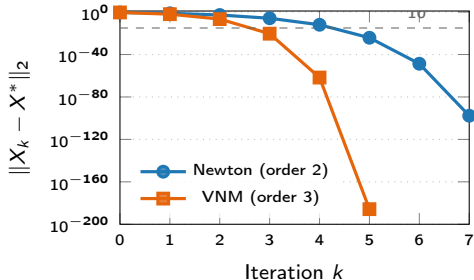


Replication results

- Iteration counts match the paper **exactly in 13/18 cases (VNM)**; within ± 1 in 17/18. VNM beats NM in **all 18**. Measured COC of VNM = 3.00 in 17/18 cases.
- Excluding the two hard starts of f_1 , totals are **almost equal** (192 vs 194 paper; 210 vs 206 ours): VNM's real edge is **robustness far from the root**.
- We found a **typo** in the paper's function (8): printed $\cos x \sin x$ must be $\cos x + \sin x$ (residual $-4.94 \rightarrow 5 \times 10^{-11}$).

6. Extension to Nonlinear Systems and Conclusions

System: $x^2 + y^2 = 4$, $xy = 1$, **start** $X_0 = (1.5, 0.5)$



Multivariate VNM

- Same idea with the Jacobian J :
 $(J(X_n) + J(Y_n))\Delta X = 2F(X_n)$.
- **3 iterations vs. 5** for Newton (**40% fewer**);
measured COC = 3.0 vs. 2.0.

Conclusions

- Base paper **confirmed**: VNM is cubic, error constant $C_2^2 + \frac{1}{2}C_3$ matches to 6 digits.
- Practical gain = **robustness**, not fewer evaluations near the root.
- **S-VNM** is cubic (not 4th); **St-VNM** is quadratic (not 3rd).
- **Future work**: true 4th order needs a new *corrector* (e.g. Ostrowski, Jarratt), not a better quadrature rule.